FINC 4101
Investment Analysis

Topic: Interest Rate Risk

Learning Objectives

1. Explain how maturity, coupon rate, and YTM affect a bond's interest-rate sensitivity.
2. Compute and interpret Macaulay duration and modified duration.
3. Use modified duration to estimate percentage and dollar changes in bond value.
4. Explain why duration is an approximation and how convexity affects its accuracy.
5. Relate a bond's duration to an investor's investment horizon and interest-rate exposure.

Interest-Rate Risk: Which Bond Moves More?

- Suppose market yields rise by 1 percentage point (100 basis points). Will every bond lose the same percentage of value?
- Assume both bonds have the same initial YTM and coupon-payment frequency, with no embedded options.
  A. 5-year, 8% coupon
  B. 30-year, 8% coupon

Interest Rate Risk

Why are changes in interest rates a risk for bonds?

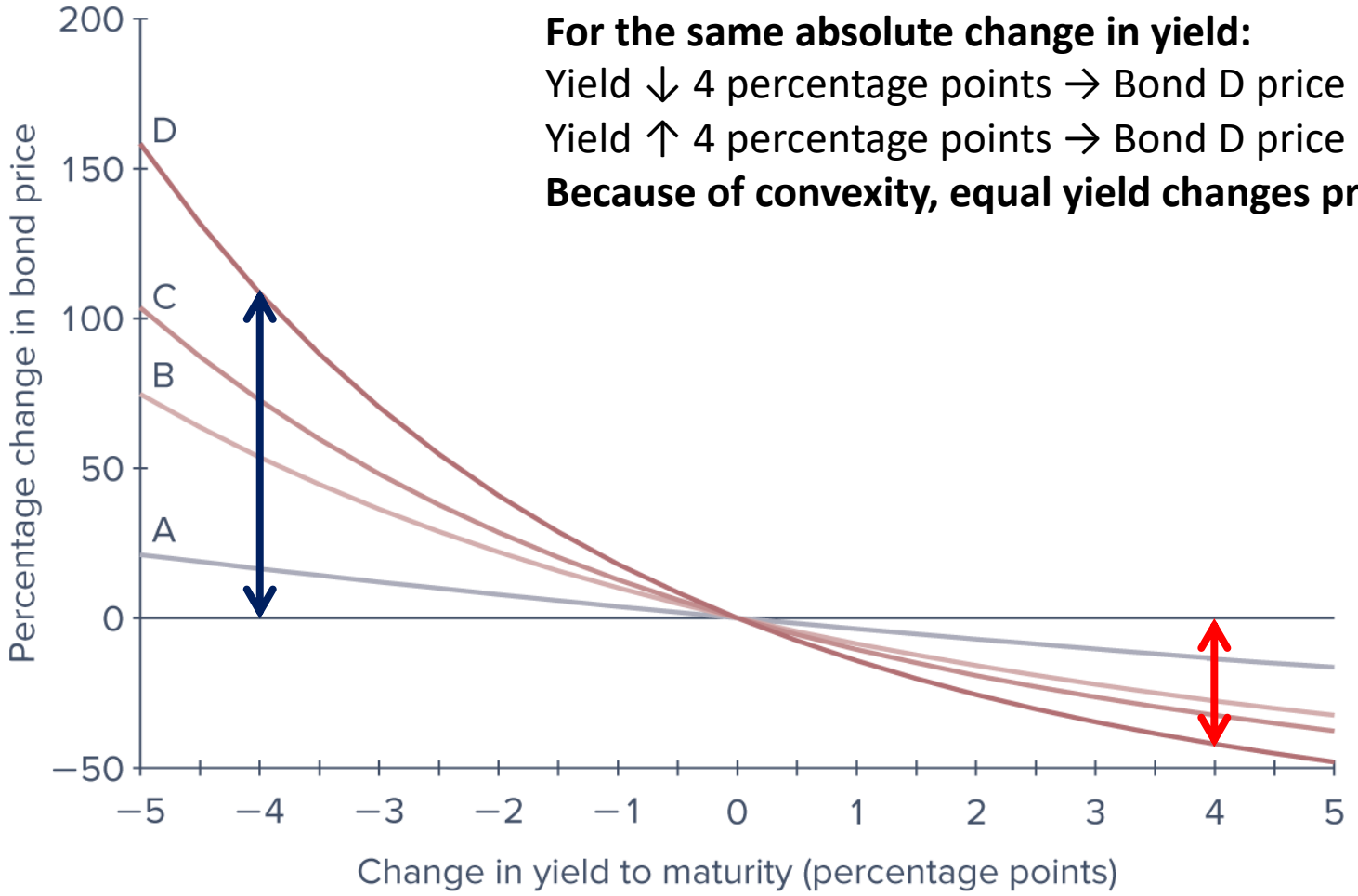
Changes in interest rates → Changes in bond price (capital gain/loss) → Uncertainty in return from bond investment

- **Interest rate risk**: sensitivity of a bond's value and returns to changes in market interest rates.
  - It primarily affects bond prices and reinvestment of coupon payments
- **Our focus in this chapter is bond price sensitivity to changes in market yields**.

Which Bond Is More Sensitive to Interest Rates?

Bond	Coupon	Maturity	Initial YTM
A	12%	5 yrs	10%
B	12%	30 yrs	10%
C	3%	30 yrs	10%
D	3%	30 yrs	6%

Bond Price Sensitivity Is Nonlinear: Convexity



For the same absolute change in yield:
Yield ↓ 4 percentage points → Bond D price ↑ about 110%
Yield ↑ 4 percentage points → Bond D price ↓ about 40%
Because of convexity, equal yield changes produce unequal price changes.

Click here to Compare Bond Price Sensitivity

Bond	Coupon	Maturity	Initial YTM
A	12%	5 years	10%
B	12	30	10
C	3	30	10
D	3	30	6

What Determines Bond Price Sensitivity?

Holding other factors constant:

- **Longer maturity → greater sensitivity**
  - Interest-rate sensitivity generally increases with maturity.
- **Lower coupon rate → greater sensitivity**
  - More of the bond's value is tied to the distant principal payment.
- **Lower YTM → greater sensitivity**
  - Distant cash flows receive relatively greater weight.

➢ Long maturity + low coupon + low YTM = greater interest-rate exposure.

From Maturity to Duration

- Maturity is a **major determinant** of interest-rate sensitivity.
- But bonds with the **same maturity** can have different sensitivities.
- Cash flows received before maturity affect a bond's effective exposure to interest-rate changes.
- We therefore need a measure that incorporates both:
  - **time to maturity**
  - **timing of cash flows**

➢ **Macaulay duration** measures this effective maturity.

Macaulay Duration: Effective Maturity

- Macaulay duration is the **weighted average time** until the bond's cash flows are received.
- Each cash flow's weight is its **present value as a percentage of the bond price**.
- Earlier cash flows shorten duration.
- Later cash flows lengthen duration.

➢ For a coupon bond, duration is generally less than maturity.

Macaulay's Duration Formula

$$D = \sum_{t=1}^{T} t \times w_t$$

where

$$w_t = \frac{CF_t / (1+y)^t}{\text{Bond Price}}$$

w_t= PV of cash flow t ÷ bond price

For **semiannual bonds**:

- Use the semiannual YTM and semiannual cash flows.
- If duration is calculated in half-year periods, **divide by 2** to express it in years.

Click here to explore Duration

Macaulay Duration Example

10% coupon | 2 years | semiannual coupons| YTM = 15% → Periodic YTM = 7.5% per half-year

	Period t	Cash Flow	PV of Cash Flow	Weight w_t	$t \times w_t$
	1	50	46.5116	$\frac{46.5116}{916.2668} = 0.0508$	0.0508
Coupon	2	50	43.2666	0.0472	0.0944
bond	3	50	40.2480	0.0439	0.1318
	4	1050	786.2406	0.8581	3.4324
Sum:			PV = 916.2668	1.0000	$\sum t \times w_t = 3.7093$ **half-years**

$$D_{Mac} = 3.7093 \text{ half-years}/2 = \boxed{\mathbf{1.8547 \text{ years}}}$$

Zero-Coupon Bond: A Useful Benchmark

2 years | semiannual periods| YTM = 15% → Periodic YTM = 7.5% per half-year

	Period t	Cash Flow	Weight $\mathbf{w}_t$	$t \times \mathbf{w}_t$
	1	0	0	0
Zero-Coupon	2	0	0	0
Bond	3	0	0	0
	4	1000	1	4

$D_{Mac} = 4$ half−years$/2 =$ **2 years**

- **10% coupon bond:** $D_{Mac} = 1.85$ years
- **Zero-coupon bond:** $D_{Mac} = 2.00$ years

➢ A zero-coupon bond's Macaulay duration equals its maturity.

From Macaulay Duration to Modified Duration

- **Macaulay duration:**
  Measures the weighted-average timing of cash flows.
- **Modified duration:**
  Translates that timing into approximate **price sensitivity**.

$$D_{mod} = \frac{D_{Mac}}{1 + \frac{YTM}{m}}$$

- m = number of coupon payments per year

Modified Duration Example

$$D_{Mac} = 1.8547, \qquad YTM = 15\%, \qquad m = 2$$

$$D_{mod} = \frac{1.8547}{1 + 0.15/2} = \boxed{\mathbf{1.7253\ years}}$$

- A modified duration of 1.725 means that a **1 percentage-point change in YTM** implies an approximate **1.725% change in price in the opposite direction**.

Using Modified Duration to Estimate ***Percentage*** Price Change

Dollar change in bond price

Change in YTM

$$\frac{\Delta P}{P} \approx -D_{mod} \times \Delta y$$

Initial bond price

- Δy = new YTM − initial YTM, expressed as a decimal.
- Example: 15% → 16% means $\Delta y = 0.01$.

Which Position Has Greater Dollar Exposure?

- Assume both positions have fixed cash flows, and their yields rise immediately by 50 basis points (0.50 percentage points), so Δy=0.005.

Position	Market Value	Modified duration (years)	Price change (%)	Value change ($)
A	$ 2 million	7		
B	$ 5 million	3		

1. Which position has the larger percentage loss?
2. Which position has the larger dollar loss?

$$\Delta P \approx P_0 \times -D_{mod} \times \Delta y$$

- P_0 = current market value

- Dollar exposure depends on both duration and position size

How Accurate Is Modified Duration?

- The percentage change in price for a 2-year, semiannual bond when YTM increases by **1 percentage point, from 15% to 16%**:

	Coupon = 10%	Coupon = 0%
Modified Duration (yrs)	1.7253	1.8605
Price at YTM= 15%	$916.27	$748.80
Price at YTM= 16%	$900.64	$735.03
Actual % price change	$\frac{900.64 - 916.27}{916.27} = -1.71\%$	$\frac{735.03 - 748.80}{748.80} = -1.84\%$
Estimated % price change	$-1.7253 \times 0.01 \approx -1.73\%$	$-1.8605 \times 0.01 \approx -1.86\%$

- Modified duration provides a close, but not exact, approximation for small yield changes.

When Is Duration Less Accurate?

- Modified duration is a **linear approximation** of a curved price-yield relationship.
- It works especially well for **small yield changes**.
- For **large yield changes**, the approximation error becomes more important.
- This is where **convexity** matters.

Large Yield Shock Example

- Consider an illustrative 10-year Treasury note:
  - Face value and initial price: **$1,000**
  - Remaining maturity: **10 years**
  - Annual coupon rate: **1.5%, paid semiannually**
  - Initial YTM: **1.5%**
  - Modified duration: **9.2540 years**
- YTM rises immediately to **5.0%**, an increase of **350 bps**.

1. Use modified duration to estimate the percentage price change and the new bond price.
2. Will duration overstate or understate the loss?

Large Yield Shock: Duration Estimate vs. Actual Price

1. Estimated Percentage Change:

$$\frac{\Delta P}{P} \approx -D_{mod} \times \Delta y \approx -9.254 \times 0.035 = -0.3239 \; or \; -32.39\%$$

2. Estimated Price:

$$P_1 \approx \$1{,}000 \times (1 - 9.254 \times 0.035) = \$676.11$$

3. Actual Price with YTM = 5% is $\$727.19$

$$\text{Duration Estimate} = -32.39\%$$

$$\text{Actual price change} = -27.28\%$$

Duration Error Grows with the Yield Change

- **Same 10-year bond: initial YTM = 1.5%, initial price = $1,000.**

New YTM	Yield increase	Duration-estimated price	Exact price
2.0%	50 bp	$953.73	$954.89
5.0%	350 bp	$676.11	$727.19

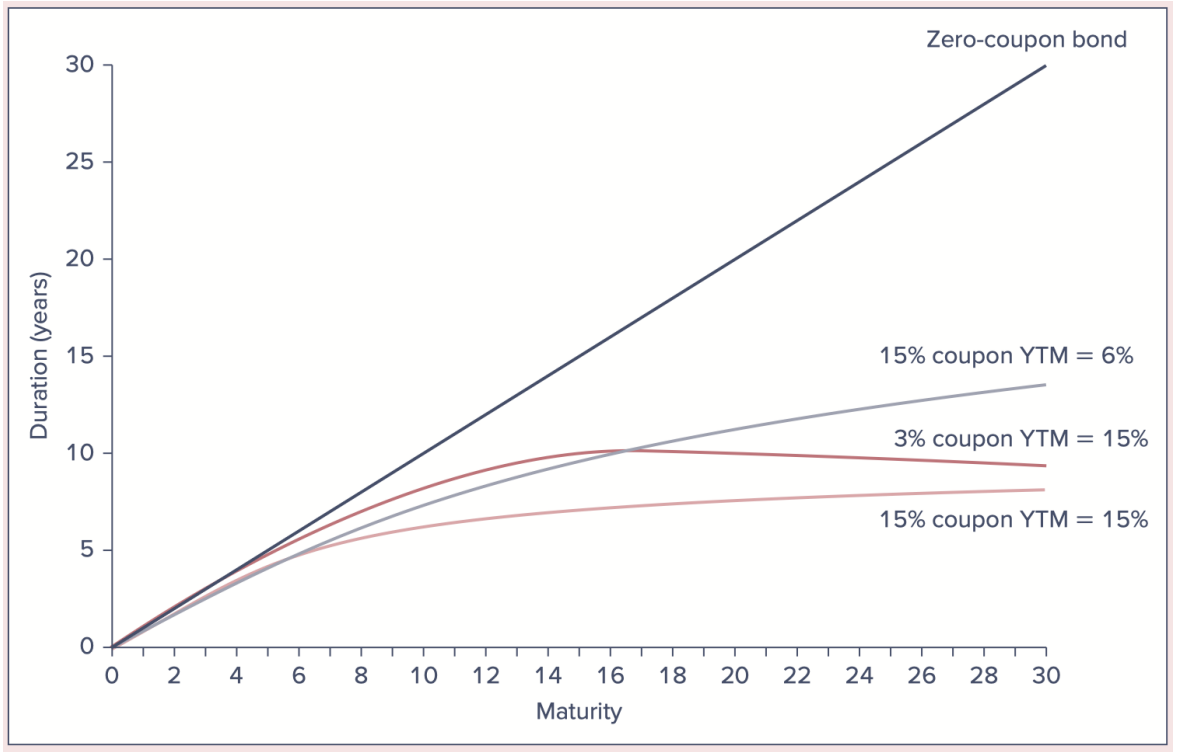
- Duration **overstates** the loss, especially for the larger yield increase.
- If yields fall instead, does duration overstate or understate the price gain?

Determinants of Macaulay Duration

Holding other factors constant:

- **Longer maturity** → generally higher duration
- **Lower coupon rate** → higher duration
- **Lower YTM** → higher duration for coupon bond
- Zero-coupon bond:

$$D_{Mac} = \text{maturity}$$



Click here to explore Duration Determinants

Ranking Bonds by Duration

- **Without calculating duration**, rank these bonds from highest to lowest Macaulay duration.

Bond	Coupon (%)	Time to Maturity (years)	Yield to Maturity (%)
A	15	20	10
B	15	15	10
C	0	20	10
D	8	20	10
E	15	15	15

Investment Horizon and Interest-Rate Risk

- Rising yields lower bond prices but increase coupon reinvestment income.

Investment horizon compared with Macaulay duration	Main exposure near the initial yield
Shorter than duration	**Price effects** tend to dominate.
Approximately equal to duration	Price and reinvestment effects **approximately offset**.
Longer than duration	**Reinvestment effects** tend to dominate.

- Interest-rate risk depends on the investor's horizon relative to Macaulay duration.

Practice Questions for Exam 2

Chapter 11:

- 4. A bond currently sells for $1,050…
- 8. a-b Find the duration of a 6% coupon…
- 9. A nine-year bond paying coupons annually…
- 12. You own a fixed-income asset with…
- 14. Long-term Treasury bonds currently are selling at yields…